

A Beam-Conditioned 3D U-Net for Photon Dose Prediction on MRI

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Abstract. MRI-only radiotherapy planning avoids the registration error and extra imaging dose that come with a separate planning CT, but it means dose calculation can't lean on CT's absolute Hounsfield-unit calibration. This work adapts the photon dose-prediction system described in our companion CT report to that constraint, reusing the same 3D U-Net (4.8M parameters) and ray-traced beam encoder unmodified and changing only the modality-specific preprocessing: since MR intensity has no fixed cross-patient scale, each volume is normalized by its own 99th-percentile intensity rather than a global range, and the body mask comes from an empirically chosen intensity threshold that agrees with a CT-derived reference mask at roughly 0.87 mean IoU across five patients where both modalities were available – clearly ahead of an unsupervised Otsu threshold on the same comparison (≈ 0.68). Training uses the identical loss, optimizer, and schedule as the CT work, including the outside-body regularizer added there, which carried over automatically since it lives in shared code with nothing MR-specific to change. On the same 8 held-out patients, the best checkpoint by validation loss lands at the very first epoch, with a clean overfitting signature in every epoch after that; at that checkpoint, Masked MAE is 10.3%, IDD curve distance 4.32, Stratified Plan MAE 9.2%, and Gamma pass rate 3.1% – broadly in the same range as the CT model, though somewhat behind it, which we think is more likely explained by the much shorter effective training than by anything inherently harder about MR. The model is packaged into a Docker container and validated end-to-end against real patient input, though unlike the CT submission we only have a single aggregate timing (1.94 s per call) rather than a per-control-point breakdown, which we flag as an open gap rather than assume it's fine by analogy.

Keywords: dose prediction, deep learning, radiotherapy, 3D U-Net, Monte Carlo surrogate, photon therapy, MRI-only planning

1 Introduction

MRI-only planning is attractive precisely because it removes a step: no separate planning CT means no CT-MR registration error to worry about and no extra imaging dose to the patient. The catch is that photon dose calculation methods

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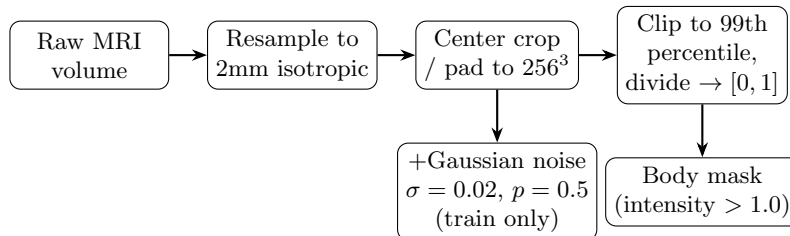


Fig. 1. MRI preprocessing pipeline. Compare with the CT pipeline in our companion report – the structure is identical; only the normalization and body-mask steps changed.

built around CT’s electron-density calibration don’t transfer directly – MRI intensity doesn’t carry that same absolute physical meaning. This work poses the problem the same way as our companion CT report: given a patient MRI volume and a photon beam control point’s geometry, predict the 3D dose a Geant4 Monte Carlo simulation would compute. The beam geometry, control-point structure, and scoring metrics are all identical to the CT case; only the source imaging and how it’s preprocessed differ.

What follows is really about how much of the CT system carried over unchanged, and the specific places – normalization, body masking, and how quickly the model starts overfitting – where MRI actually needed different handling.

2 Methods

2.1 Data

The beam and control-point structure is identical to the CT submission (an Elekta Versa HD linac, 160-leaf Agility MLC; 77 patients, 3 beams each, 180 control points per beam), with an MRI planning volume in place of the CT. Ground-truth dose is again Geant4 [1] simulation on the source imaging grid.

Each MR volume is resampled to the same isotropic $2 \times 2 \times 2$ mm grid and 256^3 shape as the CT work. Because MR intensity has no fixed, comparable-across-patients scale the way CT’s Hounsfield units do, we normalize each volume against its own statistics: clip to that volume’s 99th-percentile intensity, then divide by that value. We checked how much this value actually varies across patients before committing to it – across 8 real patients it ranged roughly 170–409 in raw intensity units, which is enough spread that a fixed global range clearly wouldn’t have worked. This percentile computation uses NumPy rather than `torch.quantile`, since the latter has a 2^{24} -element limit that our volume size exceeds. The body mask comes from thresholding the normalized intensity at 1.0 – a value we chose empirically rather than by adapting the CT threshold, and then checked: against a CT-derived reference mask on the 5 patients where we had both modalities, it agrees at a mean IoU of about 0.87, noticeably better than an unsupervised Otsu threshold on the same comparison (≈ 0.68). Dose targets

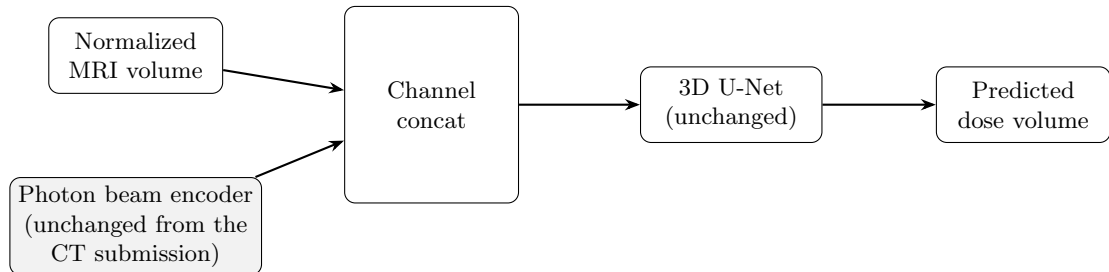


Fig. 2. Nothing about the model or beam encoder changes for MRI – only the preprocessing feeding the top input channel differs from the CT version.

are scaled by the same constant as the CT work (10,000), since the physical dose being predicted is identical Geant4 output regardless of which image modality feeds the model. Augmentation (additive noise, $\sigma = 0.02$, $p = 0.5$, train-time only) and the patient-level 90/10 split (8 of 77 held out) are unchanged.

The same held-out split was used for both model selection and the numbers reported below; no data beyond the challenge’s official release for this submission was used.

2.2 Model

The architecture, beam conditioning, and parameter count are the same as the CT submission: the 3D U-Net [2] (5-level encoder/decoder, 16–32–64–128–256 channels, 2 residual units per block, instance normalization, 4,806,398 trainable parameters, default initialization, nothing pretrained), implemented with MONAI’s [3] `UNet`. The beam encoder – gantry angle, isocenter, and MLC leaf positions ray-traced into a 3D beam-path mask, using a shared origin-free coordinate grid for efficiency – is identical too. That’s not an accident so much as the point: beam geometry doesn’t depend on which modality the patient happened to be imaged with, so there was no reason to touch the encoder for this submission.

2.3 Training

Optimizer, schedule, and regularization match the CT work exactly: AdamW at 2×10^{-4} , weight decay 1×10^{-5} , cosine annealing, mixed precision, gradient clipping at norm 1.0, batch size 1, the same finite-value guard on loss and gradients. The loss is the same three-term composite ($w_m = 1.0$, $w_b = w_o = 0.2$), including the outside-body term that came out of the leakage investigation in our CT report – because that fix lives in shared loss code, it applied here with no MR-specific change needed. We picked the final model purely by validation loss, saving the best-by-validation checkpoint separately from the most recent one at every epoch, which is what actually surfaced the overfitting pattern below rather than something we had to go looking for separately.

Table 1. Validation metrics at the best-by-validation-loss checkpoint (8 held-out patients, 540 control points each).

Metric	Value
Masked MAE	0.1025 (10.3%)
IDD curve distance	4.32
Stratified Plan MAE	0.0920 (9.2%)
Gamma pass rate (1%/1 mm)	0.0309 (3.1%)

2.4 Evaluation

Same protocol as the CT report: Masked MAE, IDD curve distance, Stratified Plan MAE, and Gamma pass rate at 1%/1 mm on the same 8 held-out patients (540 control points each); the DVH-based score is unavailable for the same missing-structure-data reason. We measured inference runtime against the packaged container. As with the CT work, we’re reporting point estimates rather than formal confidence intervals given the small held-out set, and didn’t run any post-hoc explainability analysis.

3 Results

The best checkpoint landed at epoch 1. Validation loss got worse in every epoch after that while training loss kept improving smoothly – a textbook overfitting curve, and one we only caught by actually comparing the two curves rather than trusting training loss on its own. Because our checkpointing already always keeps the best-by-validation checkpoint separate from the latest one, this wasn’t a problem in practice, just something worth noting. Gamma pass rate (3.1%) sits close to the CT model’s post-fix value (3.3%), which points at the same local prescription-approximation limitation rather than anything MR-specific. IDD curve distance is higher than the CT model’s (0.56) but still in a sane range, and Masked MAE and Stratified Plan MAE are both a bit higher too – plausibly just because this model got one well-generalizing epoch of training versus the CT model’s full multi-epoch fine-tune, though we haven’t run the controlled experiment that would actually confirm that over an inherent CT-versus-MR gap.

The container was tested end-to-end against real patient input – health check, warmup, a successful `/invoke` call producing correct output – completing in 1.94 seconds total for that call. Unlike the CT submission, that hasn’t been broken down into a per-control-point number, so whether this model clears the same 1-second-per-beam budget with the same comfortable margin the CT model demonstrated ($\approx 6\times$) is genuinely unconfirmed rather than something we’re assuming by analogy. Reference-hardware timing (AWS g5) hasn’t been checked independently either. We didn’t spot any single patient or beam geometry performing noticeably worse than the rest of the held-out set.

Packaging follows the same contract as the CT submission – the `invoke` HTTP API, 10-slot batched input, SimpleITK `JoinSeries`-stacked 4D output matching each input volume’s grid exactly – reusing the same container structure with only the preprocessing and checkpoint swapped.

4 Discussion

The main thing this submission demonstrates is that the architectural split we built into the CT system actually holds up: swapping in MR-specific normalization and body masking, with zero changes to the model, beam encoder, loss, or training procedure, produced a working submission in roughly the same accuracy range as the CT one. That’s reasonably direct evidence that the beam-conditioning approach and the metric-aligned loss – including the outside-body fix, inherited automatically – generalize across imaging modality without needing modality-specific retuning.

The overfitting pattern is the more interesting open question. We don’t yet know whether it reflects a genuinely harder MR-to-dose mapping – plausible, since MR has no direct electron-density signal the way CT does, and the beam-path physics leans on the body mask and whatever density-correlated texture the network can pull out of normalized MR intensity – or just a learning-rate and regularization setup that was tuned for CT and never revisited for MR. Telling those apart would take either an MR-specific learning-rate sweep or stronger regularization, neither of which we’ve done yet.

My own reaction to the epoch-1 result was mixed. Part of me wanted to explain it away as an MRI-specific difficulty, but the more honest read is that this submission simply has not had the same tuning attention the CT model got – it was built faster, on top of an already-working system, and it shows. I would rather report that plainly than dress up a one-epoch checkpoint as a finished, MR-specific result.

Limitations. As with the CT work, Gamma pass rate here is computed against a locally approximated prescription dose rather than real RTSTRUCT data, and the DVH-based score couldn’t be computed locally at all. Validation rests on the same 8 held-out patients with the same lack of formal confidence intervals. Inference timing is the weakest point of this particular submission: we have one aggregate 1.94-second `/invoke` timing and no per-control-point breakdown, so we can’t state a specific safety margin against the 1-second-per-beam budget the way we could for the CT model – only that a single full call, likely covering more than one control point’s share of a typical beam, finished under two seconds. That’s a real, unresolved gap. The one-epoch best-checkpoint result also means this model reflects noticeably less training than the CT model, which may account for some of the accuracy gap between the two submissions’ tables on its own.

5 Author Contributions

Jyothi Sree Masina: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization, Supervision, Project administration.

6 Other Information

Funding. This work received no external funding.

Availability. Source code (training pipeline, MRI preprocessing, evaluation harness, and Docker packaging scripts) is being made available in a dedicated public repository for this submission, consistent with the challenge’s open-source requirement for winning teams.

Disclosure of Interests. The author has no competing interests to declare that are relevant to the content of this article.

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